Pathfinder Software Requirements Specification (SRS)

Overview

Project Name: Pathfinder
Team Members:

- Patrick King - Project Manager
- Maddie McDowell - Front-End and UI Design
- Nic Ortiz - Front-End
- Thomas Powell - Back-End

Brief Description:
Pathfinder is an all-in-one career development web application designed to guide users through every stage of their professional and educational journey. It offers AI-powered career assistance through resume analysis, automated job matching, and networking resources. The platform leverages OpenAI for natural language processing, Firebase for data management, and modern web technologies for a seamless user experience.

Project Scope

This document details the software requirements for Pathfinder, a web application developed using React, HTML, CSS, and Node.js. The target users include high school and college students, job seekers, and professionals seeking career transitions. The primary goal of Pathfinder is to provide users with a streamlined career development experience by integrating multiple tools into a single platform. Features include AI-assisted career guidance, interactive resume-building tools, and seamless job search capabilities.

Component Breakdown

Front-End

- **Landing Page:** Login/Sign-Up functionality with responsive design for various devices.
- **Dashboard:** Displays personalized user information and provides access to core features.
- **Chatbot UI:** AI chatbot accessible on all pages for quick assistance.
- **Profile Setup Page:** Allows users to enter and edit personal and career preference details.

Back-End

- **Authentication & User Management:** Handles login and registration using Firebase.
- **Career Guidance Engine:** Processes user data to provide career recommendations.
- **Resume & Job Matching System:** Integrates AI-powered resume analysis and automated job matching.
- **API Integration Layer:** Connects with external services for job listings and learning resources.

AI-Powered Chatbot

- **Query Processing:** Routes user queries to OpenAI API for response generation.
- **Context Management:** Retains session context for a conversational experience.

Database & Data Storage

- **Firebase Database:** Stores user profiles, preferences, and historical data securely.

Deployment

- Hosted on cloud platforms (e.g., AWS or Google Cloud) for scalability and reliability.

Required States and Modes

- **Landing Page State:** Initial page for logging in or signing up.
- **Dashboard State:** Centralized hub displaying user-specific tools and information.
- **Settings State:** Allows customization of account settings and preferences.

Functional Requirements

User Authentication & Profile Management

- **FR1:** The website shall support user authentication via Firebase.
- **FR2:** Users shall be able to create and manage career preference profiles.

Career Guidance & Job Matching

- **FR3:** The website shall provide personalized career suggestions based on a user questionnaire.
- **FR4:** The system shall match jobs based on user-selected career paths.
- **FR5:** The job-matching system shall refresh results every 24 hours.
- **FR6:** The job-matching system shall allow users to filter results based on salary, location, and remote work availability.

- **FR7:** The system shall recommend upskilling courses based on job listings that do not fully match user qualifications.

Resume & Cover Letter Assistance

- **FR8:** The website shall analyze uploaded resumes and suggest improvements in specific areas.
- **FR9:** The resume analysis tool shall provide suggestions specific to job descriptions uploaded by the user.
- **FR10:** The system shall generate customizable cover letter templates tailored to job applications.

AI Chatbot Assistance

- **FR11:** The website shall include an AI chatbot assistant for career-related guidance.
- **FR12:** The chatbot shall be accessible through a floating button in the bottom-right corner of all pages.
- **FR13:** The chatbot shall interact with users via a text-based interface with an optional voice input feature.

Networking & Learning Resources

- **FR14:** The system shall integrate with LinkedIn API or a similar platform to fetch networking and learning opportunities.
- **FR15:** The platform shall suggest networking events and mentorship programs based on user preferences.

User Engagement & Notifications

- **FR16:** Users shall receive notifications when new job opportunities, career resources, or networking events align with their profile.
- **FR17:** The dashboard shall display weekly career development tips and personalized action items.

External Interface Requirements

- **EIR1:** Integrate with job board APIs for fetching relevant job listings.
- **EIR2:** Use Firebase for secure data storage and authentication.

Internal Data Requirements

- **IDR1:** User data, including profile and preferences, shall be securely stored in a Firebase database.

- **IDR2:** Questionnaire results and career path data shall be retrievable for real-time user interactions.

Adaptation Requirements

- **AR1:** The website shall adapt seamlessly to varying screen sizes, ensuring usability on mobile, tablet, and desktop devices.

Security Requirements

- **SecR1:** User data, such as email and passwords, shall be encrypted during transmission and storage.
- **SecR2:** Only authenticated users shall have access to personalized features.

Environmental Requirements

- **ER1:** The application shall support major browsers, including Chrome, Firefox, Safari, and Edge.

Computer Resource Requirements

- **CRR1:** The system shall perform optimally on low-spec mobile devices.

Software Quality Factors

- **SQF1:** The interface shall be intuitive, minimizing the need for external user guidance.
- **SQF2:** Downtime for maintenance shall be communicated in advance to users.

Design & Implementation Constraints

- **DIC1:** The development process shall be constrained by the capabilities of React, Node.js, Firebase, and OpenAI APIs.

Training-related Requirements

- **TRR1:** A built-in tutorial shall guide users through the application's features during the initial setup.

Other Requirements

- **OR1:** Accessibility features such as screen reader compatibility and keyboard navigation shall be implemented.